ULAB
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Counting Rules

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The Fundamental Counting Rule

Fundamental Counting Rule

In a sequence of n events in which the first one has k_1 possibilities and the second event has k_2 and the third has k_3, and so forth, the total number of possibilities of the sequence will be

$$k_1 \cdot k_2 \cdot k_3 \cdots k_n$$

Note: In this case *and* means to multiply.

মৌলিক গণনা সূত্র (Fundamental Counting Rule)

যদি কোনো ধারাবাহিক nটি ঘটনা ঘটে, এবং

- প্রথম ঘটনার সম্ভাব্য উপায়ের সংখ্যা হয় k_1,
- দ্বিতীয় ঘটনার সম্ভাব্য উপায়ের সংখ্যা হয় k_2,
- তৃতীয় ঘটনার সম্ভাব্য উপায়ের সংখ্যা হয় k_3,
  এভাবে চলতে থাকে,

তাহলে পুরো ক্রমটির সম্ভাব্য মোট উপায়ের সংখ্যা হবে —

$$k_1 \times k_2 \times k_3 \times \cdots \times k_n$$

দ্রষ্টব্য: এখানে “এবং” (and) শব্দটি গুণ (multiply) বোঝায়।

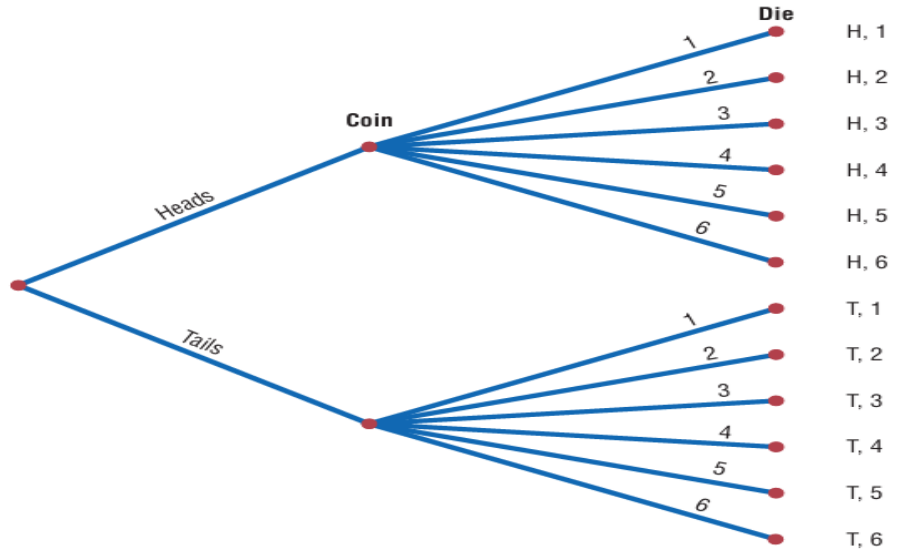
EXAMPLE 4–38 Tossing a Coin and Rolling a Die

A coin is tossed and a die is rolled. Find the number of outcomes for the sequence of events.

SOLUTION

Since the coin can land either heads up or tails up and since the die can land with any one of six numbers showing face up, there are $2 \cdot 6 = 12$ possibilities. A tree diagram can also be drawn for the sequence of events. See Figure 4–10.

FIGURE 4–10
Complete Tree Diagram for Example 4–38



EXAMPLE 4–41 Railroad Memorial License Plates

The first year the state of Pennsylvania issued railroad memorial license plates, the plates had a picture of a steam engine followed by four digits. Assuming that repetitions are allowed, how many railroad memorial plates could be issued?

SOLUTION

Since there are four spaces to fill for each space, the total number of plates that can be issued is $10 \cdot 10 \cdot 10 \cdot 10 = 10{,}000$. *Note:* Actually there was such a demand for the plates, Pennsylvania had to use letters also.

EXAMPLE 4–40 Distribution of Blood Types

There are four blood types, A, B, AB, and O. Blood can also be Rh+ and Rh−. Finally, a blood donor can be classified as either male or female. How many different ways can a donor have his or her blood labeled?

SOLUTION

Since there are 4 possibilities for blood type, 2 possibilities for Rh factor, and 2 possibilities for the gender of the donor, there are $4 \cdot 2 \cdot 2$, or 16, different classification categories, as shown.

Blood type		Rh		Gender	
$\boxed{4}$	$\cdot$	$\boxed{2}$	$\cdot$	$\boxed{2}$	$= 16$

Combinations

A selection of distinct objects without regard to order is called a **combination.**

The number of combinations of r objects selected from n objects is denoted by nCr and is given by the formula:

$$_nC_r = \frac{n!}{(n-r)!r!}$$

EXAMPLE 4–48 Flier Types

An advertising executive must select 3 different photographs for an advertising flier. If she has 10 different photographs that can be used, how many ways can she select 3 of them?

SOLUTION

$$_{10}C_3 = \frac{10!}{(10-3)!3!} = \frac{10!}{7!3!} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$

Hence, there are 120 different ways to select 3 photographs from 10 photographs.

EXAMPLE 4–49 Committee Selection

In a club there are 7 women and 5 men. A committee of 3 women and 2 men is to be chosen. How many different possibilities are there?

SOLUTION

Here, you must select 3 women from 7 women, which can be done in ${}_7C_3$, or 35, ways. Next, 2 men must be selected from 5 men, which can be done in ${}_5C_2$, or 10, ways. Finally, by the fundamental counting rule, the total number of different ways is $35 \cdot 10 = 350$, since you are choosing both men and women. Using the formula gives

$${}_7C_3 \cdot {}_5C_2 = \frac{7!}{(7-3)!3!} \cdot \frac{5!}{(5-2)!2!} = 350$$

EXAMPLE 4–50 Four Aces

Find the probability of getting 4 aces when 5 cards are drawn from an ordinary deck of cards.

SOLUTION

There are ${}_{52}C_5$ ways to draw 5 cards from a deck. There is only 1 way to get 4 aces (that is, ${}_4C_4$), but there are 48 possibilities to get the fifth card. Therefore, there are 48 ways to get 4 aces and 1 other card. Hence,

$$P(4 \text{ aces}) = \frac{{}_4C_4 \cdot 48}{{}_{52}C_5} = \frac{1 \cdot 48}{2{,}598{,}960} = \frac{48}{2{,}598{,}960} = \frac{1}{54{,}145}$$

EXAMPLE 4–51 Defective Transistors

A box contains 24 transistors, 4 of which are defective. If 4 are sold at random, find the following probabilities.

a. Exactly 2 are defective.

b. None is defective.

c. All are defective.

d. At least 1 is defective.

SOLUTION

There are ${}_{24}C_4$ ways to sell 4 transistors, so the denominator in each case will be 10,626.

a. Two defective transistors can be selected as ${}_4C_2$ and two nondefective ones as ${}_{20}C_2$. Hence,

$$P(\text{exactly 2 defectives}) = \frac{{}_4C_2 \cdot {}_{20}C_2}{{}_{24}C_4} = \frac{1140}{10{,}626} = \frac{190}{1771}$$

b. The number of ways to choose no defectives is ${}_{20}C_4$. Hence,

$$P(\text{no defectives}) = \frac{{}_{20}C_4}{{}_{24}C_4} = \frac{4845}{10{,}626} = \frac{1615}{3542}$$

c. The number of ways to choose 4 defectives from 4 is ${}_4C_4$, or 1. Hence,

$$P(\text{all defective}) = \frac{1}{{}_{24}C_4} = \frac{1}{10{,}626}$$

d. To find the probability of at least 1 defective transistor, find the probability that there are no defective transistors, and then subtract that probability from 1.

$$\begin{aligned} P(\text{at least 1 defective}) &= 1 - P(\text{no defectives}) \\ &= 1 - \frac{{}_{20}C_4}{{}_{24}C_4} = 1 - \frac{1615}{3542} = \frac{1927}{3542} \end{aligned}$$

EXAMPLE 4–53 State Lottery Number

In the Pennsylvania State Lottery, a person selects a three-digit number and repetitions are permitted. If a winning number is selected, find the probability that it will have all three digits the same.

SOLUTION

Since there are 10 different digits, there are $10 \cdot 10 \cdot 10 = 1000$ ways to select a winning number. When all of the digits are the same, that is, 000, 111, 222, . . . , 999, there are 10 possibilities, so the probability of selecting a winning number which has 3 identical digits is $\frac{10}{1000} = \frac{1}{100}$.

EXAMPLE 4–54 Tennis Tournament

There are 8 married couples in a tennis club. If 1 man and 1 woman are selected at random to plan the summer tournament, find the probability that they are married to each other.

SOLUTION

Since there are 8 ways to select the man and 8 ways to select the woman, there are $8 \cdot 8$, or 64, ways to select 1 man and 1 woman. Since there are 8 married couples, the solution is $\frac{8}{64} = \frac{1}{8}$.

11. Cable Television In 2006, 86% of U.S. households had cable TV. Choose 3 households at random. Find the probability that

a. None of the 3 households had cable TV

b. All 3 households had cable TV

c. At least 1 of the 3 households had cable TV

10. Selecting Marbles A bag contains 9 red marbles, 8 white marbles, and 6 blue marbles. Randomly choose two marbles, one at a time, and without replacement. Find the following.

a. The probability that the first marble is red and the second is white

b. The probability that both are the same color

c. The probability that the second marble is blue

16. Winning a Door Prize At a gathering consisting of 10 men and 20 women, two door prizes are awarded. Find the probability that both prizes are won by men. The winning ticket is not replaced. Would you consider this event likely or unlikely to occur?

32. Doctor Specialties Below are listed the numbers of doctors in various specialties by gender.

	Pathology	Pediatrics	Psychiatry
Male	12,575	33,020	27,803
Female	5,604	33,351	12,292

Choose one doctor at random.

a. Find P (male|pediatrician).

b. Find P (pathologist|female).

c. Are the characteristics "female" and "pathologist" independent? Explain.

6. Defective Resistors A package contains 12 resistors, 3 of which are defective. If 4 are selected, find the probability of getting

a. 0 defective resistors

b. 1 defective resistor

c. 3 defective resistors